Pilot Guidebook Test Procedure

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1 Pilot Guidebook Validation Test Procedure

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Document Information

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Position Name	**Field or Pilot Team**	**Name**
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2 Test Objective

The objective of this test procedure is to evaluate the usability, completeness, and effectiveness of the Pilot Guidebook in guiding a pilot through operation of the Damage Assessment Protocol (D.A.P.) system.

This test verifies that a pilot can use the Pilot Guidebook to:

- Understand the purpose of the D.A.P. system.
- Execute the D.A.P. workflow using mission `.log` and `.nc` files.
- Generate a Final Damage Assessment Report.
- Interpret the Primary Diagnosis and fault severity outputs.
- Understand how the Digital Twin and Diagnoser contribute to the final diagnosis.

Four mission datasets are used throughout this validation test. Each mission dataset contains one `.log` file and one `.nc` file. The pilot can be be provided the expected diagnosis prior to execution.

This test validates the Pilot Guidebook as an operational documentation product. This test does not evaluate the internal diagnostic accuracy of the D.A.P. system itself.

The outputs generated during this test include pilot-recorded diagnoses, pilot interpretation results, workflow completion status, assistance records, and pilot feedback regarding guidebook usability.

3 Test Procedure

3.1 Equipment

Table 1: Equipment Required for Pilot Guidebook Validation Test

Item	Description	Value
Seaglider Piloting Laptop or Workstation with MATLAB	Must have MATLAB R2024b or later installed; must be able to run `.m` scripts and load mission files; no calibration required.	1
`coefficientsV4_25wing.m`	Flight coefficients for 25% wing Seaglider.	1
`coefficientsV4_50wing.m`	Flight coefficients for 50% wing Seaglider.	1
`coefficientsV4_50rudder.m`	Flight coefficients for 50% rudder Seaglider.	1
`coefficientsV4_nominal.m`	Flight coefficients for nominal Seaglider.	1
`Create_Mission_Data_Matrix.m`	Compiles relevant mission data into matrix form for analysis.	1
`D_fault_isolator.m`	Isolates fault based on data analysis.	1
`DAP_Main.m`	Main file used to execute the D.A.P. system.	1
`Digital_Twin.m`	Simulates Seaglider states based on fault cases and compiles data into matrix form.	1
`dynamicsV7.m`	Simulates Seaglider kinematic states.	1
`Log_File_Unpacker.m`	Unpacks relevant data from log files.	1
`missioncompare_detector.m`	Compares mission data to simulated Digital Twin data to determine whether the error bound is exceeded.	1
`NC_File_Unpacker.m`	Unpacks relevant data from NC files.	1
`parametersV5.m`	Sets parameters used by the dynamics code.	1
Pilot Guidebook PDF	Documentation being validated during this test.	1
Mission Datasets	Contains P1940008, P1940009, P1950701, and P1950702 `.log` and `.nc` files.	1
Validation Recording Sheet	Used by the evaluator to record pilot performance, diagnoses, interpretation accuracy, and feedback.	1

3.2 Pilot Guidebook Test Setup

Before beginning Pilot Guidebook validation testing, the following setup steps must be completed:

1. ______ Turn on Seaglider Piloting Laptop or workstation.
2. ______ Open MATLAB.

3. ______ Confirm MATLAB version by typing `ver` into the command window.

   Is MATLAB version R2024b or later? ______

4. ______ Confirm that the D.A.P. System folder contains the required files listed in Table 1.

   All required files present? ______

5. ______ Confirm the Pilot Guidebook PDF is available.

   Pilot Guidebook available? ______

6. ______ Confirm the following mission datasets are available:

   - P1940008 `.log` and `.nc`
   - P1940009 `.log` and `.nc`
   - P1950701 `.log` and `.nc`
   - P1950702 `.log` and `.nc`

   All datasets present? ______

7. ______ Confirm the validation recording sheet is available.

   Recording sheet available? ______

8. ______ Open `DAP_Main.m` in MATLAB.

   `DAP_Main.m` opened? ______

The tester should now be ready to begin Pilot Guidebook validation testing.

3.3 Pilot Guidebook Validation Procedure

3.3.1 Pilot Guidebook Validation Workflow

The following procedure shall be completed for each mission dataset.

1. ______ Direct the pilot to review Section 1: System Overview of the Pilot Guidebook.

   System Overview reviewed? ________________

2. ______ Identify the mission dataset being evaluated.

   Mission dataset confirmed? ________________

3. ______ Using Section 2: D.A.P. Manual of the Pilot Guidebook, execute the D.A.P. system.

   D.A.P. execution started? ________________

4. ______ When prompted for the `.log` file, select the corresponding mission `.log` file.

   `.log` file selected? __________________

5. ______ When prompted for the `.nc` file, select the corresponding mission `.nc` file.

   `.nc` file selected? __________________

6. ______ Allow the D.A.P. system to complete execution.

   Code executing? __________________

   Final Damage Assessment Report generated? __________________

7. ______ Record the Primary Diagnosis.

   Primary Diagnosis recorded? __________________

8. ______ Record the reported fault severity.

   Fault severity recorded? __________________

9. ______ Using Section 3: D.A.P. Output and Interpretation of the Pilot Guidebook, explain the meaning of the Primary Diagnosis and reported fault severity.

   Output successfully interpreted? __________________

10. ______ Using Section 4: Understanding D.A.P. Results of the Pilot Guidebook, explain how the D.A.P. system generated the diagnosis.

    Mission Data Matrix understood? ______

    Digital Twin output comparison understood? ______

    Percent error calculation understood? ______

    Allowable error margins understood? ______

    Window score calculation understood? ______

    Fault threshold understood? ______

    Sustained failure logic understood? ______

    Fault isolation process understood? ______

    Fault confidence / average percent error relationship understood? ______

    Final diagnosis generation understood? ______

11. ______ Record whether pilot assistance was required.

    Assistance required? __________________

12. ______ Repeat Steps 1–11 for all remaining mission datasets.

The following mission datasets shall be evaluated:

Table 2: Pilot Guidebook Validation Mission Datasets

Dataset	**Files Required**	**Expected Diagnosis**
P1940008	`.log` and `.nc`	25% Wing Fault
P1940009	`.log` and `.nc`	Nominal Case
P1950701	`.log` and `.nc`	50% Rudder Fault
P1950702	`.log` and `.nc`	50% Wing Fault

All datasets completed? ______

3.4 Shutdown

1. ______ Save all recorded validation results and pilot feedback.

   Files archived? ______

2. ______ Close MATLAB.

   MATLAB closed? ______

3. ______ Shut down Seaglider Piloting Laptop or workstation.

   Laptop shutdown complete? ______

4 Acceptance Criteria

The following Yes/No acceptance criteria are recorded for each Pilot Guidebook validation test:

1. (Yes/No) Pilot successfully executed the D.A.P. workflow.
2. (Yes/No) Pilot generated a Final Damage Assessment Report.
3. (Yes/No) Pilot successfully recorded the Primary Diagnosis.
4. (Yes/No) Pilot successfully interpreted the D.A.P. outputs.
5. (Yes/No) Pilot successfully explained how the diagnosis was generated using the Pilot Guidebook.
6. (Yes/No) Pilot completed all four mission datasets.
7. (Yes/No) Pilot completed the workflow without developer assistance.
8. (Yes/No) Pilot provided usability feedback for the Pilot Guidebook.

5 Version History

Version	Date	Author	Description of Changes
1.0	May 3, 2026	EP	Initial Draft
1.1	May 8, 2026	EP	Edited to include Pilot Guidebook validation workflow and acceptance criteria
1.2	May 13, 2026	EP	Edited to include different mission datasets and expected diagnoses for validation testing
1.3	May 18, 2026	EP	Edited to include Pilot Guidebook updates and revised D.A.P. Manual workflow
1.4	May 22, 2026	EP	Edited to include Pilot Guidebook Output Interpretation section validation steps
1.5	May 26, 2026	EP	Edited to include Understanding D.A.P. Results section validation, fault confidence interpretation, window score interpretation, and percent error evaluation
1.6	May 29, 2026	EP	Finalized test procedure formatting and validation workflow

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